Heatmap_Final

Greg Fisher

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```
library(tidyverse)
library(ComplexHeatmap)
library(circlize)
```

```
count.df <- read.csv(file = "data/counts.csv", header = TRUE,row.names = 1)
meta_data.df <- read.csv(file = "data/meta_data.csv", header = TRUE)
```

```
# Recommended 10-gene core BRCA/HR variant panel For mutation/variant analysis

genes.of.interest <- c("ENSG00000012048.23","ENSG00000139618.16",
                       "ENSG00000083093.10","ENSG00000138376.11",
                       "ENSG00000108384.15", "ENSG00000185379.20",
                       "ENSG00000136492.9", "ENSG00000149311.19",
                       "ENSG00000183765.23", "ENSG00000167258.15")

count.df.subset <- count.df %>%
  filter(rownames(.) %in% genes.of.interest)

count.df.matrix <- as.matrix(count.df.subset[,])

# Rename from Ensebml ID to Gene
rownames(count.df.matrix) <- c("BRCA1", "BRCA2", "PALB2", "BARD1", "RAD51C",
                               "RAD51D", "BRIP1", "ATM", "CHEK2", "CDK12")
```

```
# Number of samples to choose in column space of count matrix
n <- 30

count.df.matrix.subset <- count.df.matrix[,1:n]

# Number of rows of the metadata need to match the number of columns of the count matrix
meta_data.df.subset <- meta_data.df[1:n,c(1,10,23,25,26, 27,29)]
```

```
# Define your target strings
targets <- c("Adjuvant", "Anastrozole", "First-Line Therapy", "Carboplatin","Paclitaxel")
pattern <- paste(targets, collapse = "|")

# Extract the matching string into a new column
meta_data.final.df <- meta_data.df.subset %>%
  mutate(treatment_method = str_extract(treatments, pattern))
```

```
# Define color schemes for all annotations
colour_list <- list(
  laterality = colorRamp2(c(18, 45, 90), c("Left" = "blue", "Right" = "orange", "NA" = "yellow")),
  sample_type = colorRamp2(c(18, 90), c("Primary Tumor" = "red","Solid Tissue Normal" = "green")),
```

```
  #sample_type = colorRamp2(c(20, 100), c("red","green")),
  treatment_method = c("Adjuvant" = "white", "Anastrozole" = "lightblue","Carboplatin" = "lightpink", "
)
```

```
# Create top annotation for continuous variables
top_ann <- HeatmapAnnotation(
  Laterality = meta_data.final.df$laterality,
  Sample_type = meta_data.final.df$sample_type,

  col = colour_list,
     na_col = "blue",
     gp = gpar(col = "green"),

  annotation_name_gp = gpar(fontsize = 10),
  annotation_legend_param = list(
    Laterality = list(title = "Laterality"),
    Sample_type = list(title = "Sample Type")
  )
)

# Create bottom annotation for treatment variables
bottom_ann <- HeatmapAnnotation(
  Treatment_Method = meta_data.final.df$treatment_method,

  col = colour_list,
     na_col = "blue",
     gp = gpar(col = "white"),

  annotation_name_gp = gpar(fontsize = 10)

)
```

```
# Heatmap with annotations
annotated_heatmap <- Heatmap(
  count.df.matrix.subset,
  name = "Expression",

  # Clustering parameters
  clustering_distance_rows = "spearman",
  clustering_method_rows = "ward.D2",
  clustering_distance_columns = "spearman",
  clustering_method_columns = "ward.D2",

  #col = colour_list,

  top_annotation = top_ann,
  bottom_annotation = bottom_ann,

  show_row_names = TRUE,
  row_names_gp = gpar(fontsize = 8),
  show_column_names = TRUE,
  column_names_gp = gpar(fontsize = 8),

  column_title = "Gene Expression"
```

```
)
```

```
## The automatically generated colors map from the 1^st and 99^th of the
## values in the matrix. There are outliers in the matrix whose patterns
## might be hidden by this color mapping. You can manually set the color
## to `col` argument.
##
## Use `suppressMessages()` to turn off this message.
```

```
draw(annotated_heatmap)
```

